

Statistics Hybrid Practice Set – S13 (L9)

Topics Covered

- Population & Sample
- Mean, Median, Mode
- Range, Variance, Standard Deviation
- Quartiles & IQR
- Outlier Detection
- Frequency Table
- Histogram
- **Percentiles (New)** ★
- Density Plot interpretation

Instructions

- **First solve on paper.**
- Then solve the **same question in Excel.**
- No formula-recall questions.
- For interpretation questions, explain **why**.

Q1. Population & Sample

A company has **12,000 employees**. It selects **500 employees** to study employee satisfaction.

Paper: Identify the population and sample.

Excel: Enter the two values in a small table.

Q2. Mean vs Median

The following are monthly delivery times in minutes:

Delivery Time
18
20
21
22
24
25
80

Paper

1. Calculate the **Mean**.
2. Calculate the **Median**.
3. Which one better represents the typical delivery time?
4. Why?

Excel

Calculate the same values and verify your answer.

Q3. Dispersion Comparison

Two delivery teams have the following delivery times.

Team A

20, 21, 22, 23, 24

Team B

15, 18, 22, 26, 29

Paper

Calculate the **Range** for both teams.

Then answer:

1. Which team has greater spread?
2. Which team appears more consistent?

Excel

Calculate the range for both teams.

★ Q4. Percentile – New Concept

The following are test scores:

40, 45, 50, 55, 60, 65, 70, 75, 80, 85

A student scored **75**.

Paper

Without using Excel first, determine approximately what position 75 occupies in this ordered dataset.

Then answer:

Is 75 a relatively low, middle, or high score compared with the other students?

Excel

Use Excel's percentile function to find the **75th percentile** of the dataset.

Then compare the result with the score **75**.

★ Q5. Percentile Interpretation

A company's employee performance score has a **90th percentile value of 86**.

What does this mean?

- A. 90% of employees scored exactly 86.
- B. 90% of employees scored at or below 86.
- C. 10% of employees scored at or below 86.
- D. The average score is 86.

Explain your choice in one sentence.

Q6. Quartiles + Percentiles

The following dataset represents customer spending:

100, 120, 140, 160, 180, 200, 220, 240, 260, 280

Calculate:

1. Q1
2. Q2
3. Q3
4. IQR
5. 25th percentile
6. 75th percentile

Interpretation

What relationship do you notice between:

- Q1 and the 25th percentile?
- Q3 and the 75th percentile?

Excel

Verify all calculations.

Q7. Outlier + Percentile Scenario

A company's delivery times are:

10, 11, 12, 13, 14, 15, 16, 17, 50

Answer:

1. Is **50** likely to be an outlier?
2. Which method would you use to confirm it?
3. Would the **mean or median** better represent typical delivery time?
4. Why can an extreme value affect the mean?

Excel

Calculate:

- Mean
 - Median
 - Q1
 - Q3
 - IQR
-

Q8. Frequency Distribution + Percentile Interpretation

A school groups students according to their marks:

Marks	Students
0–19	2
20–39	5
40–59	8
60–79	12
80–100	3

Answer:

1. Which mark range has the highest frequency?
2. How many students scored **60 or above**?
3. How many students are in total?
4. Which range contains the largest concentration of students?

Excel

Create the same frequency table.

Q9. Real Data Analyst Scenario

A company wants to identify the employees whose performance scores are among the **top 10%** of all employees.

Which concept would be most useful?

- A. Range
- B. Percentile
- C. Mode
- D. Variance

Explain why.

Q10. Mixed Business Case

An online store records customer spending:

₹500, ₹520, ₹550, ₹580, ₹600, ₹620, ₹650, ₹700, ₹2,500

Answer:

1. Is there likely an outlier?
2. Which method would you use to confirm it?
3. Would you use Mean or Median to represent typical customer spending?
4. If a customer is at the **90th percentile**, what does that tell you about their spending compared with other customers?
5. Why can percentiles be useful to a Data Analyst?